Lesson 2.1

Patterns & Inductive Reasoning

Look for the pattern, then predict

What is inductive reasoning?

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1. Look $\rightarrow$ **2.** Describe the rule $\rightarrow$ **3.** Predict

A number pattern

3, 7, 11, 15, ...

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Rule: **add 4** each time (check every pair: $7-3$, $11-7$, $15-11$)

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10th term: $3+9\cdot 4=39$ start at 3, add 4 nine times

A picture pattern



1, 4, 9, 16, ...

A picture pattern



1, 4, 9, 16, ...

These are **square numbers**: $1^2, 2^2, 3^2, 4^2$. Next: $5^2 = 25$

Conjecture and counterexample

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Counterexample: $-2 + 5 = 3$, and $3 < 5$. So the conjecture is **false**.

Watch out

2, 4, 8, ... times 2? $\rightarrow$ 16

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2, 4, 8, ... times 2? $\rightarrow 16$

Watch out! Or $+2, +4, +6$? $\rightarrow 14$. Two rules can fit the same first terms — that's why inductive reasoning gives a **conjecture**, not a proof.

Your turn! Pause the video and try it.

Find the next two terms: $1,\ 3,\ 6,\ 10,\ \ldots$

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Find the next two terms: $1,\ 3,\ 6,\ 10,\ \ldots$

The differences are $+2, +3, +4$, so next $+5$, then $+6$:

$10 + 5 = 15$, $15 + 6 = 21$

Remember

- **Inductive reasoning**: look, describe the rule, predict.
- Check the rule on **every** pair of terms.
- A pattern gives a **conjecture**; one **counterexample** shows it is false.

Now practice at your level:

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